

# Computational Modeling

*Object-Based Attention · model write-up · generated from open-perception.org*

Valdés-Sosa et al. (2000) introduced a transparent-motion design that has spawned numerous subsequent psychophysical and neurophysiological studies of surface-based attention. Taken together, these studies have been argued to provide some of the best evidence for object- or surface-based selection (Reynolds, Alborzian, & Stoner, 2003; Çatak, Özkan, Kafalıgönül, & Stoner, 2022).

We are implementing computational models to provide a framework for understanding experimental findings from these previous studies and to provide predictions for future experiments.

Background: what are we trying to model?

Valdés-Sosa, Cobo, and Pinilla (2000) introduced a transparent-motion design as a tool to examine object-based attention absent the potential influence of feature-based or spatial attention. There have been numerous follow-up studies that have used variants of that design to investigate behavioral and/or neuronal correlates of object-based attention. Although there have been some design differences, all of these studies used stimuli composed of two superimposed counter-rotating, differently colored dot fields. One of the two dot fields is “cued”, either endogenously (e.g., fixation point color indicating the color of field to be attended) or exogenously (e.g., by a delayed-onset of one dot field, see below). The rotations of the dot fields are interrupted by brief translations (one or two translations, depending upon the design), and subjects are asked to report the direction of those translations. These translations consist of a subset of the dots moving coherently in typically one of eight directions. To discourage tracking of individual dots, the remaining dots of the translating dot field translate in randomly chosen directions. Numerous studies using this basic design have repeatedly found that subjects judge translations of the cued dot field more accurately than translations of the uncued dot field (Khoe et al., 2005; Lopez et al., 2004; Mitchell et al., 2003; Mitchell, Stoner, & Reynolds, 2004; Pinilla et al., 2001; Reynolds et al., 2003; Rodriguez et al., 2002; Stoner & Blanc, 2010; Valdés-Sosa, Cobo, & Pinilla, 2000). Several of these studies also reported electrophysiological (ERP) correlates of the cueing advantage (Valdés-Sosa, Bobes, Rodriguez, & Pinilla, 1998; Pinilla et al., 2001; López, Rodriguez, & Valdés-Sosa, 2004; Rodríguez & Valdés-Sosa, 2006; Çatak et al., 2022); in the delayed-onset design specifically, Çatak et al. (2022) is the only study to date to have reported ERP effects.

As observed by Stoner & Blanc (2010), for the object-based account to hold, the attentional cue — whether endogenous, exogenous, or both — must first privilege the subset of overlapping dots rotating in one direction and then privilege those same dots when they translate. The visual system must therefore “bind” the cued rotation to the subsequent translation in an object-specific manner. The mechanism of such binding has never been identified, nor has any directly applicable model been advanced. In this section, we implement a series of computational models, of varying complexity, to provide a framework for understanding experimental findings from previous studies and to provide predictions for future experiments. We begin with a non-object-based alternative — one that must be evaluated before the object-specific machinery of binding can be meaningfully addressed.

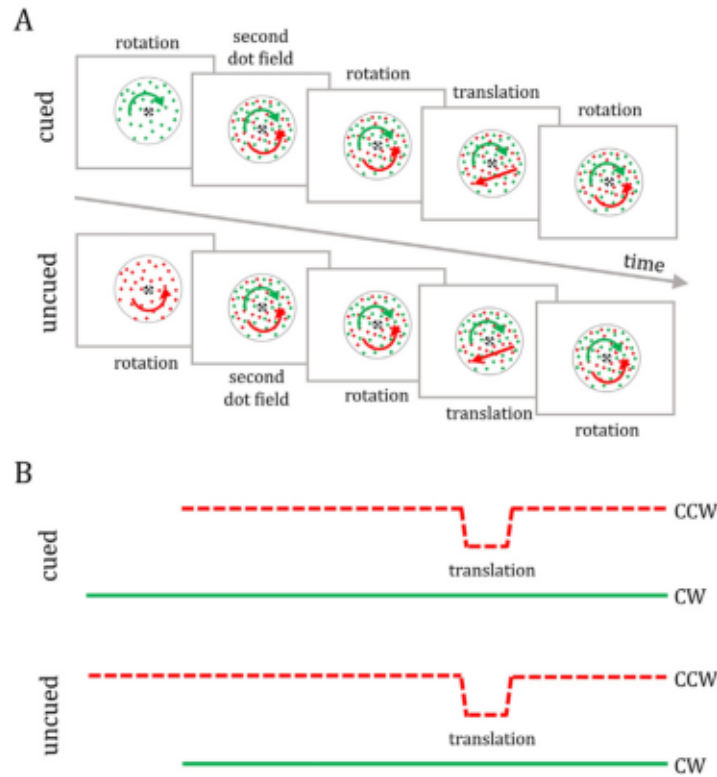


Figure 1. Delayed-onset design (Çatak et al., 2022). (A) One rotating dot field appears followed by the second "delayed" dot field. Two superimposed dot fields rotate in opposite directions around the fixation point, allowing the perception of two transparent surfaces. Following the rotation, either the delayed (cued) or non-delayed (uncued) dot field translates briefly. After the translation, both dot fields continue to rotate. (B) Feature-based illustration of timeline. The two dot fields are differentiated by line style (dashed or solid) and with dot field colors indicated by the line colors. The vertical line placement indicates the different motion directions: clockwise rotation (CW), counter-clockwise rotation (CCW), and translation. The onset differences in this design result in "cued" translations occurring in the presence of the older rotation direction and "uncued" translations occurring in the presence of the newer rotation direction.

Figure and caption reproduced from Çatak, Özkan, Kafaligönül & Stoner (2022), *Cortex* 151, 89-104, © Elsevier.

## The motion-competition model

*Stoner & Blanc (2010) — a biased-competition circuit*

### Motion competition — a non-object-based account

In considering the possible neuronal mechanisms underlying the "object-based" effects observed in transparent-motion stimuli, we will (initially at least) focus on the delayed-onset design (Figure 1) — the single-translation variant introduced by Reynolds, Alborzian, & Stoner (2003). As reviewed above, translation directions of cued (delayed) dot fields are reported more accurately than those of the uncued dot field. Similarly, translations of the cued dot field yield larger ERPs than those of the uncued dot field (Çatak et al., 2022).

Contrary to the object-based account, Stoner & Blanc suggested that the critical issue was not whether the translation was of the cued or uncued dot field but whether it occurred in the presence of the cued or uncued rotation. Stoner & Blanc's suggested reframing of the paradigm follows from established competitive interactions among direction-selective neurons (Qian & Andersen, 1994; Snowden, Treue, Erickson, & Andersen, 1991; Krekelberg & Albright, 2005). According to the biased-

competition account of attention (Desimone & Duncan, 1995), co-activated populations of neurons compete, and the winner can be tipped by stimulus strength — such as luminance contrast — or by exogenously- or endogenously-directed selective attention (Luck, Chelazzi, Hillyard, & Desimone, 1997; Moran & Desimone, 1985; Reynolds, Chelazzi, & Desimone, 1999).

Stoner & Blanc assumed that the delayed rotation had a processing advantage, either because the delayed onset acted as an exogenous cue (Yantis & Jonides, 1984) and/or because of short-term neuronal adaptation — the decrease in neuronal response following stimulus onset observed in motion-selective neurons of areas MT (Priebe, Churchland, & Lisberger, 2002; Priebe & Lisberger, 2002) and MST (Duffy & Wurtz, 1997). Given this assumption and competitive interactions, they suggested that translations of cued dot fields have an advantage because they compete with "weaker" stimuli than do translations of uncued dot fields.

## ## Biased Competition

Stoner & Blanc (2010) applied the competition model of Reynolds, Chelazzi, & Desimone (1999) to this paradigm. That design carries a motion-duration confound: a cued translation occurs alongside the older, longer-present rotation, an uncued translation alongside the newer one. Cueing is therefore confounded with how long the surrounding rotation has been visible — a bias that requires no object-specific binding.

Two assumptions turn that confound into the cueing advantage. First, rotation-tuned neurons adapt — their response wanes the longer a direction has been present. Second, they divisively suppress a translation-tuned detector. Because a cued translation is accompanied by the older, more-adapted rotation, that rotation suppresses the detector less, and the cued translation evokes the larger response (see Stoner & Blanc, 2010). The figures below trace this computation and its sharp prediction: decouple duration from cueing with a motion swap, and the advantage should reverse.

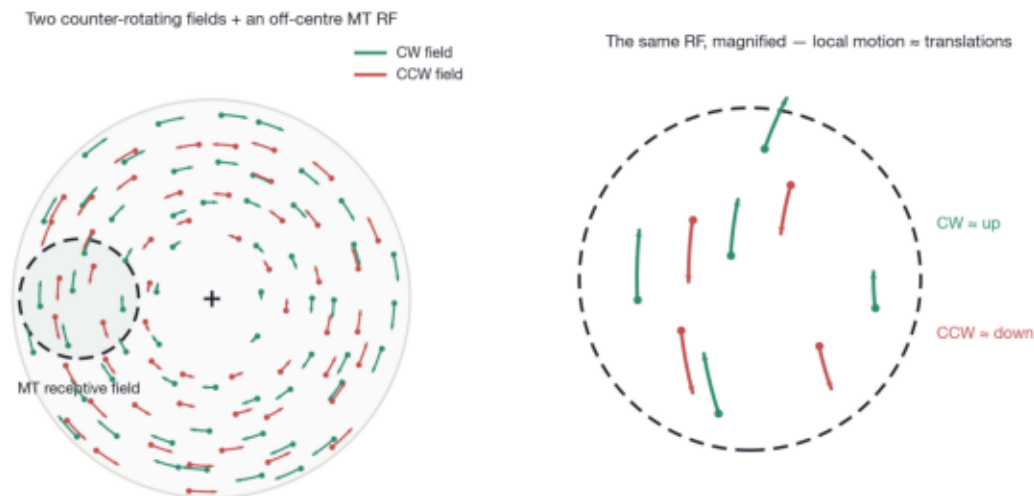


Figure 2. Why we can treat the inputs as translations. MT neurons have large receptive fields that take in both transparent surfaces at once. For a receptive field placed off-center (here directly left of fixation), each rigidly rotating field is, locally, approximately a translation: the clockwise field drifts up and the counter-clockwise field drifts down. The right panel is that same receptive field magnified — the identical dots and their identical rotation arcs, not an idealized patch — so the motions shown are exactly those in the stimulus. (Each trace is 100 ms of motion at the stimulus's 81°/s rotation; dot density is Stoner & Blanc's 5 dots/deg<sup>2</sup> per field.) So the two rotations and the brief translation can all be represented as motion directions on a single axis — the population drive that feeds the model. V1 supplies that directional signal — its small receptive fields extract the local

motion direction most precisely — but V1 is the input, not a model stage: this rotation drive depends on motion direction alone, so it is unchanged by a color or identity swap (the motion swap is taken up in Figure 3). The model's own stages live in MT. Crucially, the model does not depend on this local-translation simplification. It simply lets us reason with a directional MT drive and defer MST and rotation-selective neurons, which can be layered in later.

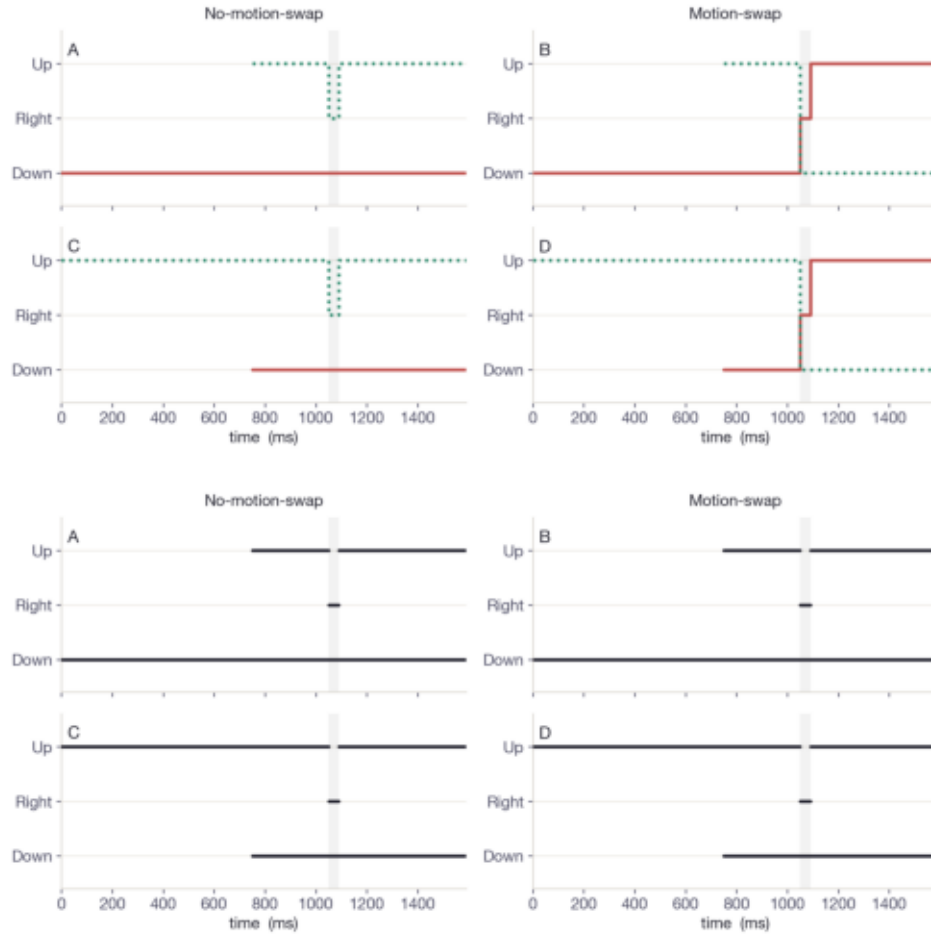


Figure 3. Feature trajectories  $\leftrightarrow$  what the model sees. Toggle (or let it alternate) between two views of the same four conditions (A-D, after Stoner & Blanc, 2010, Fig. 4). Here we take the concrete case of a single MT hypercolumn whose collective receptive field sits on the left of the rotating display: there, clockwise rotation is locally upward motion, counter-clockwise rotation is locally downward motion, and the coherent shift is a rightward translation — so the three direction channels read as Up / Right / Down. Feature trajectories: each field's motion direction over time — color = the field's dot color, line style = field identity (dotted = the field that briefly translates), with vertical jumps marking transitions. What the model sees: the very same panels with identity discarded — every line now black, solid, and reduced to the motion directions present (no color, no identity, no transitions). That stripping is exactly the abstraction the motion-competition model performs: it pools over which dots carry each direction. Once you do, A and B become identical, as do C and D— the motion swap reproduces the opposite cue's no-swap input; it only relabels which dots translate (cued  $\leftrightarrow$  uncued), leaving the directional input unchanged. Because the model sees only the second view, it cannot tell A from B or C from D, and is forced to predict that the cued advantage reverses under a motion swap — the prediction the behavioral data refute. The gray band marks the 40 ms translation; the brief onset offset ( $\approx 750$  ms) is the delayed onset of the second field.

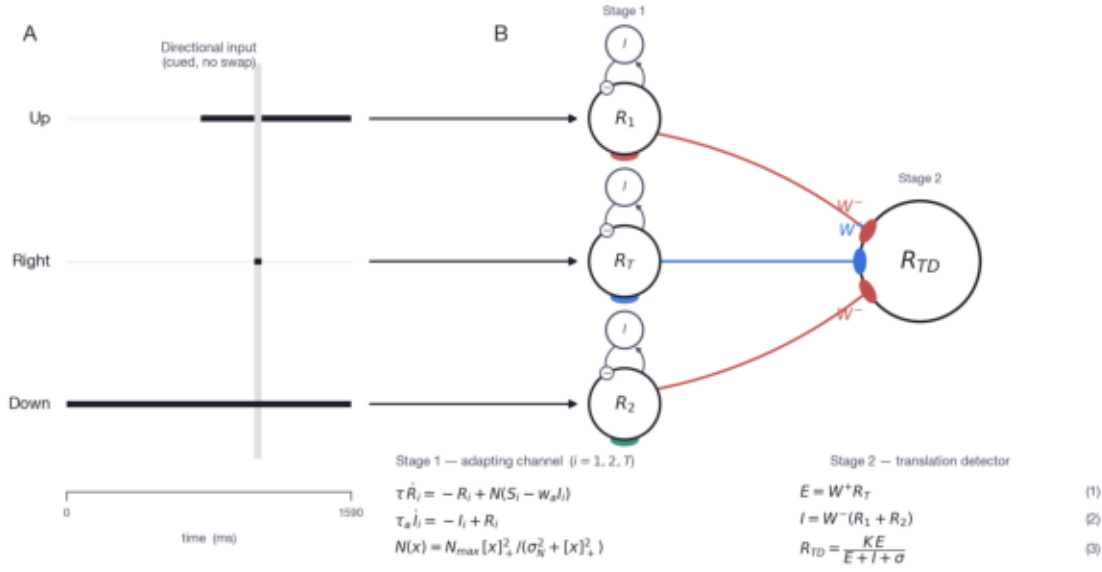


Figure 4. Inputs feeding the biased-competition circuit (Reynolds, Chelazzi & Desimone, 1999). (A) The directional input to a single left-positioned MT hypercolumn over one trial (cued, no swap): the three motion-direction channels — Up (clockwise rotation), Right (the coherent translation), and Down (counter-clockwise rotation)— drawn as the drive each receives over time, with the brief gray band marking the 40 ms translation. Each channel projects (horizontal arrow) to its Stage-1 neuron in (B), the same circuit rotated so the channels form a vertical column that lines up with the inputs: Up → R1, Right → R(T), Down → R2. The translation detector is a single neuron in MT / MST — a region whose large receptive fields take in both transparent surfaces at once. It receives excitatory drive (+) from the translation input T, and divisive inhibition (–), via local interneurons, from the two rotation channels R1 and R2 (direction-selective inputs supplied by V1→MT). The inhibitory terminals are drawn tangent to the soma in the usual way for shunting/divisive inhibition. Its steady-state response is  $R_{TD} = KE / (E + I + \sigma)$ , with  $E = W^+T$  and  $I = W^-(R_1 + R_2)$  — exactly Eq. 4 of Reynolds et al., the same biased-competition computation recorded in V2/V4 and shown to operate in MT/MST (Recanzone et al., 1997; Treue & Maunsell, 1996). The rotating fields are the competitors that suppress the detector; what biases the competition is either the adaptation difference between R1 and R2 (Stoner & Blanc, 2010 — our current model) or attention (Reynolds & Heeger, 2009).

Where the stages live. Stage 1 — the adapting rotation channels R1, R2 — sits in area MT: their receptive fields have to be large enough to take in many dots at once, so each is driven continuously by the dot flow and can therefore adapt over ~1 s. We leave Stage 2 (the translation detector) unspecified— a different layer of MT, or recurrent connections within it, would serve equally well. V1 is not a model stage but the upstream source of the local motion-direction signal. Because the channels' adaptation depends only on motion direction and onset timing— not dot color or identity — the model predicts that a color/identity swap leaves cueing intact, whereas a motion swap, which moves the less-adapted direction onto the other dots, should reverse the cued advantage— the prediction tested in Figure 3.

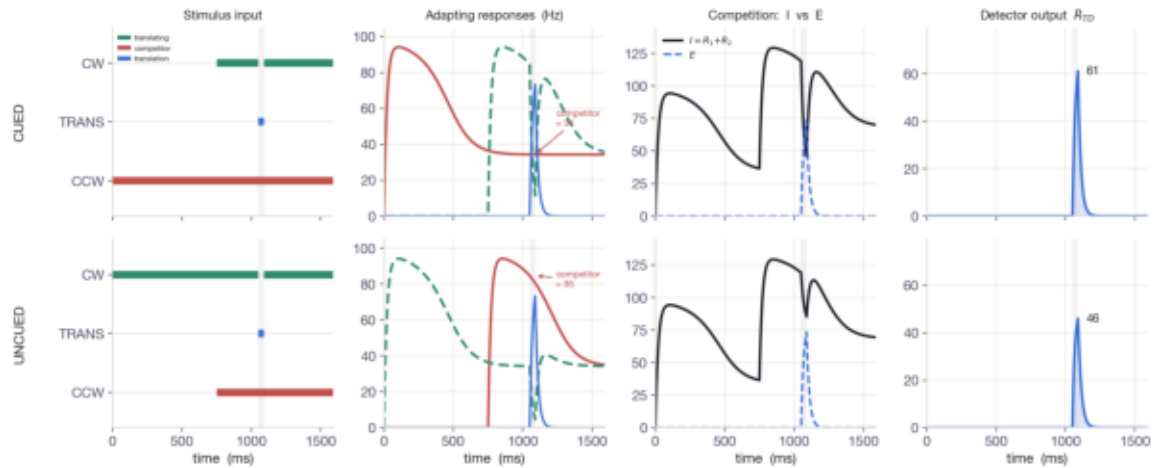


Figure 5. The computation, end to end. Each row is one of the two cases the four trial types collapse to (CUED = the delayed-onset field translates; UNCUEd = the first-on field translates — recall  $A \equiv B$  and  $C \equiv D$  at the input). Reading left to right: (1) Stimulus input — the two rotation directions and the brief translation as binary direction channels (green = the field that will translate, its rotation interrupted during the window; red = the steady competitor; blue = the translation). (2) Adapting responses — the Stage-1 MT direction channels (Eqs 1–3); the delayed field, having had less time to adapt, responds more strongly than the first-on field. (3) Competition — total divisive inhibition  $I = R_1 + R_2$  versus excitation  $E$  from the translation. During the translation the translating field's rotation is interrupted, so its channel drops out of the inhibitory pool and  $I$  notches down; the notch is deeper when the interrupted field is the less-adapted (delayed) one. (4) Detector output  $R_{TD} = KE/(E+I+\sigma)$  —  $E$  is the same in both rows, so the deeper inhibition notch in CUED yields the higher peak (here 61 vs 46, a +33% bias). Note the directional input (column 1) is identical with or without a feature swap — the swap never enters this cascade.

## The normalization model of attention

*Reynolds & Heeger (2009) — attention supplies the bias*

The competition model produced the cued advantage from adaptation, with no attentional term. Reynolds & Heeger (2009) describe a normalization model of the same biased-competition family in which the bias instead comes from attention: an attention field multiplicatively scales the neurons tuned to the attended stimulus, before divisive normalization. Running that model on the very same delayed-onset stimulus — with a fixed attentional gain on the cued direction standing in for Stoner & Blanc's adaptation — reproduces the cued advantage. The two accounts are, at this level, interchangeable: whether the cueing bias is supplied by adaptation or by attention, the normalization stage turns it into the same response difference.



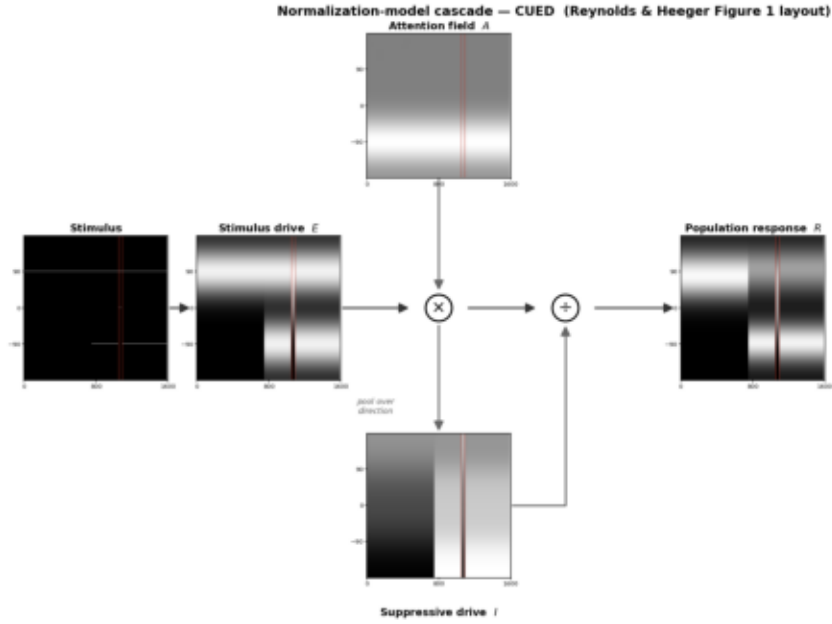


Figure 6. The normalization model on the same stimulus (Reynolds & Heeger Figure 1 layout). R&H's model written as their Figure-1 visual equation, computed with a bit-for-bit verified port of their attentionModel.m: the stimulus drives a direction-tuned stimulus drive  $E$ , which is multiplied ( $\times$ ) by the attention field  $A\theta$  — a fixed gain bump on the cued (Down /  $-90^\circ$ ) direction — and the product  $E \cdot A$  is then divided by the suppressive drive  $I$  to give the population response  $R = (E \cdot A) / (I + \sigma)$ . The suppressive drive is that same product, pooled —  $I = (E \cdot A) * (I_x \times I_\theta)$  — so the attention field shapes both the numerator and (through the pool) the denominator. Each panel is a grayscale map over preferred motion direction (vertical) and time (horizontal); white = high (following R&H, where midgray = 1 and white > 1). The suppressive drive is present throughout the trial — wherever the rotations drive the population — not a translation-only term. The attention field replaces Stoner & Blanc's adaptation as the source of bias. Axes, kernels, and normalization are R&H's, with time substituted for their spatial receptive-field axis ( $\sigma = 10^{-6}$ ) — see the note below Figure 8 on exactly what that substitution does and does not change.

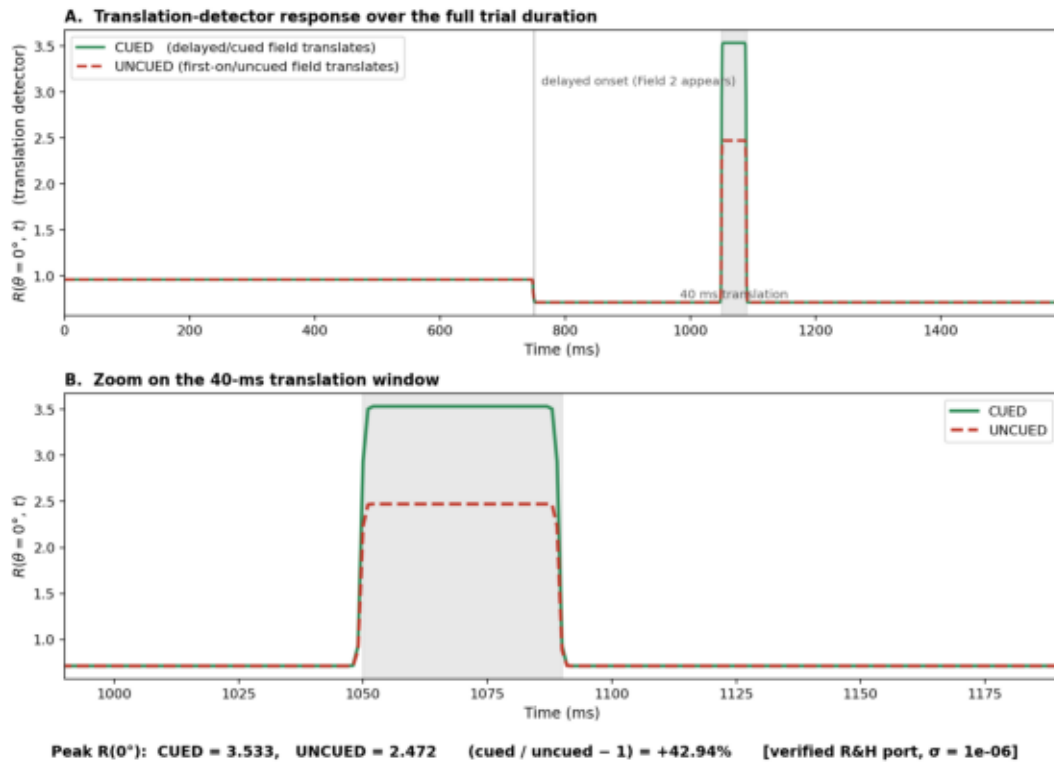


Figure 7. Attention reproduces the cued advantage. The translation-detector output  $R(\theta = 0^\circ, t)$  — the population response at the translation direction — across the whole trial (top) and zoomed on the 40 ms translation window (bottom), for CUED (green) and UNCUE (red), computed with the verified R&H port. With a single fixed attentional gain favoring the cued direction and no adaptation, the normalization model peaks higher for the cued field: CUED = 3.53 vs UNCUE = 2.47, a +43% bias (at R&H's default  $\sigma = 10^{-6}$ ). The competition model (Figure 5) produced this same advantage from adaptation alone — so at the level of this prediction the two accounts are interchangeable: the cueing bias can be supplied by adaptation or by attention. (The magnitude depends on the normalization constant  $\sigma$ ; the effect is strongest in the normalization-dominated regime and shrinks as  $\sigma$  grows.)



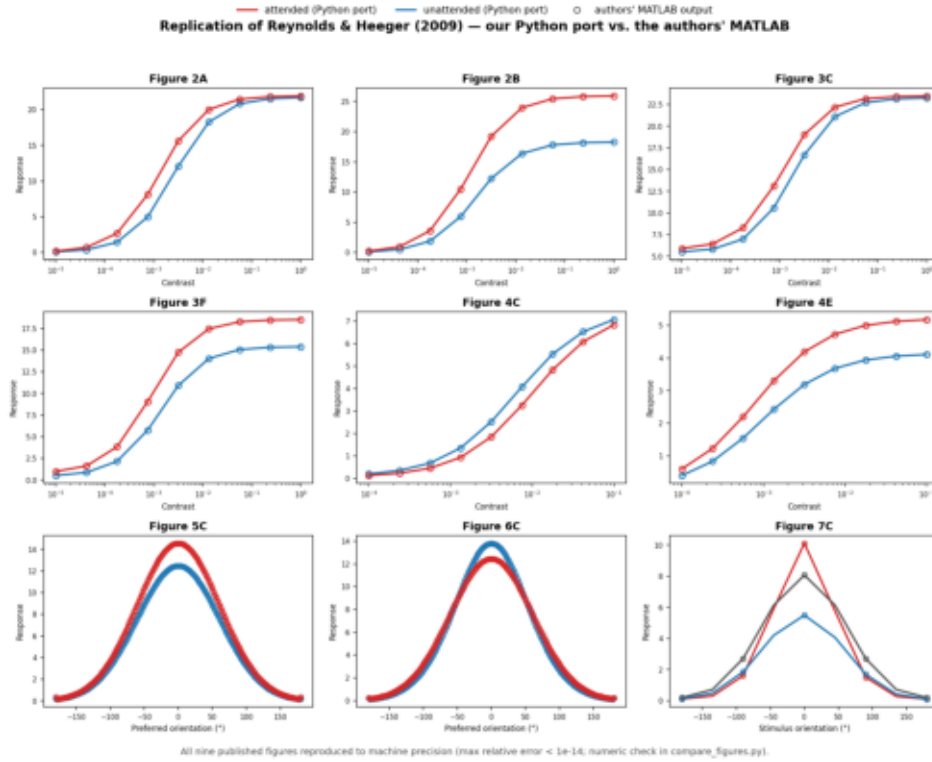


Figure 8. Our port reproduces all nine Reynolds & Heeger (2009) figures. Each panel re-runs one published case through our Python port of `attentionModel.m` (solid lines—attended in red, unattended in blue) and overlays the authors' own MATLAB output (open circles). Figures 2A-4E are contrast-response functions (response vs. log contrast, attention toward vs. away from the receptive field); 5C-6C are spatial/feature tuning curves; 7C is the orientation-variation result. The lines pass through the circles in every panel—agreement is to machine precision (max relative error  $< 10^{-14}$ ; the numeric check is in `compare_figures.py`).

How the time-varying application differs from R&H's original. The model we verified above is defined over  $\text{space} \times \text{feature}$ : the stimulus drive is pooled across neighbouring receptive-field positions and across orientation, and each of R&H's figures is a steady-state response to a static display. To drive it with our dynamic stimulus we substitute time for their spatial (receptive-field-centre) axis, so the maps in Figures 6–7 run over  $\text{direction} \times \text{time}$ . Two consequences follow, and both matter:

- (1) The normalization pool does not pool over time. We set the pooling width along the time axis to  $\approx 0$  (an impulse), so the suppressive drive at each instant is built only from the population at that instant — pooled over direction, not over time. Adjacent time frames do not normalize one another, whereas in R&H adjacent positions do. So R&H's spatial surround-suppression has no temporal analogue here; we keep only the cross-direction (feature) normalization.
- (2) There are no normalization dynamics. R&H's responses are steady-state; we apply that steady-state solution independently at every frame, i.e. we assume the normalization settles instantaneously. There is no time constant and no temporal memory.

Is it a different model? No — it is the same model in a degenerate regime. The equations are unchanged ( $E = A \cdot E_{\text{raw}}$ ,  $I = E * \text{kernel}$ ,  $R = E / (I + \sigma)$ ); we have only relabelled one axis and zeroed the pooling along it. Our application therefore reduces to R&H's feature-domain normalization solved separately at each moment, with no coupling across time.

The implication is sharp. Because nothing is carried across time, the cued advantage in Figures 6–7

is produced entirely by the fixed attention field at the translation instant — not by the delayed-onset history. (When the translating field's rotation pauses during the translation, the drive removed from the suppressive pool is large if that field was attention-boosted — the cued case — and small otherwise; the onset asynchrony itself never enters the computation.) This is the contrast with the competition model, where the same advantage arises from adaptation — a genuine temporal memory of the onset asynchrony. As applied here, the normalization account reproduces the bias but does not use the timing; it requires attention to already be pointed at the cued field.